The courtyard as the family's haven · a study in phases · Phase 0: the facts · Taif / Palm House V10.0

The owner's brief (10 Nov 2026): breeze, water, tree, shade, winter sun, views, privacy, calm, beauty, easy upkeep, all working together; find every mistake. **Locked:** the site and the outline, the structure, the stair and the lift, the settled rooms, the malqaf L5. **Open:** the seating, the water and the tree, the outlet (A or B), the sink window, the courtyard's doors and windows, the arch infill, the lighting, the floor, the planting, the drainage. **Owner's preferences:** no TV; every seat with a back, 5–6 seats; open to the sky; level floor, movable furniture; render and paint, porcelain floors; simple switches; privacy only from the guest route and the majlis side. **Amber = an assumption.** Phase 0 cites the files; Phase 1 checks them.

The courtyard as designed (V10.0 + the owner's review of 10 Nov 2026)

Item	Fact	Where
Courtyard on axes	an L: x 4.25–8.95 × y 6.5–13.0 less the malqaf core (x 4.4–5.8, y 6.5–8.667): **27.19 m²**	data/rooms_gf.csv 2
Courtyard clear (finished faces)	4.50 × 6.30 = 28.35 m² less the core's footprint (1.6 × 2.17 = 3.47): ≈ **24.88 m²**; the colonnade under the projection 1.575 wide (x 4.35–5.925), the open sky 3.15 × 6.5 on axes	courtyard_layout.typ 30–38 · sections.typ pj (depth 1.55, sky 3.15)
Walls round it	north: the women's majlis glass (+2.9 head) under the family living's W2; south: the living glass to +3.32 under W3; east: the kitchen wall two storeys to the roof (+7.3) and the 1.10 guard; west: the projection's soffit +3.27 on the colonnade, the gallery's W1 above (crown +6.60), the roof +7.3	params.yaml 11, 130 · structure.typ (soffit +3.27)
Malqaf chamber	the core x 4.4–5.8 × y 6.5–8.667 on axes (faces x 4.3–5.95, y 6.4–8.767; the east wall 0.30 on the column line since 10 Nov 2026); inside: the dust trap, the filter cabinet + EC fan + silencer, the cooler's space; the door 1.0 × 2.4 south onto the covered strip	air_system.typ 25, 31–34 · design_gf.typ 80, 138
Outlet A (as built)	1.4 × 1.0 on the east face, y 7.1–8.5, +0.1…+1.1, adjustable timber louvres aimed south-east (the air study: "the louvres aim the jet along the seating"); free area 0.84 m²; ≈ 1.79 m³/s and 2.1 m/s at the louvres at the summer-evening mean wind, 0.85 m³/s in a gentle wind, 0.49 m³/s in calm (stack)	air_system.typ 30, 50–51, 67, 228
Outlet B (studied, not adopted)	the same 1.4 × 1.0, raised to +0.60…+1.60, centred in the bay y 6.9–8.3, the louvres fixed 30° south-east with splayed reveals: over the	malqaf_outlet_B.typ 36
Seating (10 Nov 2026)	the level zone x 5.925–8.85 × y 7.1–12.3 (12.09 m² free of the enclosure): sofa A 1.5 × 0.85 facing W, sofa B 1.4 × 0.85 facing N, two armchairs 0.75 at the colonnade edge facing E, a 0.7 × 0.5 table; 6 seats, every seat with a back; no TV	courtyard_layout.typ 46–52 · design_gf.typ 79
Colonnade and arches	the edge beam on x 5.8 (C21 at y 6.5, C22 at 13.0; the first arcade column at 8.667 absorbed in the core, the second at 10.833, Ø0.25 rendered); three semicircular arch plates under the soffit +3.27, springing ≥ +2.31; the bays 2.167 on axes; the first bay filled by the core (the outlet in it)	character.yaml 79–81 · structure.typ X18 · design_gf.typ 74–80
Projection and gallery	the first-floor projection 1.55 m over the colonnade (B3 / B4 on 4.33 + 6.25 m spans); the gallery behind W1 (sill +3.82), the family living behind W2, the living's upper side W3; the sky opening 3.15 wide between the edge beam and the kitchen wall	sections.typ 38–53 · structure.typ 73–74
Drains	the floor −0.015 at the three glass lines and the riwaq door, 1 % falls to −0.06 at the kitchen wall foot; two slot drains (along the kitchen wall foot y 9.6–12.8; at the riwaq door y 9.0–10.6), silt baskets, trap primers; the overflow gully at (8.6, 12.6) rim −0.03; the invert −0.55, 1 % east under the kitchen to the rain chamber (never foul)	character.yaml 153–162 · plumbing.typ 79, 328
Lighting	two hanging lanterns under the soffit edge between the columns (at y 9.75 and 16.125, 0.6 m drops); the warm room lights seen through the glass (the night view); nothing else designed	character.yaml 170–173, 202–205
Power and water points	the pump in the enclosure's divider (its circuit not drawn); the TV's power / data removed (10 Nov 2026); no sockets or lighting circuit for the courtyard designed yet; the malqaf chamber: IP44 sockets, a capped water point, the gully	courtyard_layout.typ 187 · air_system.typ § 2
Floor	outdoor anti-slip porcelain (R11) in a sand / honey tone, no pattern designed; level, no step;	DESIGN_LANGUAGE.md · character.yaml 153

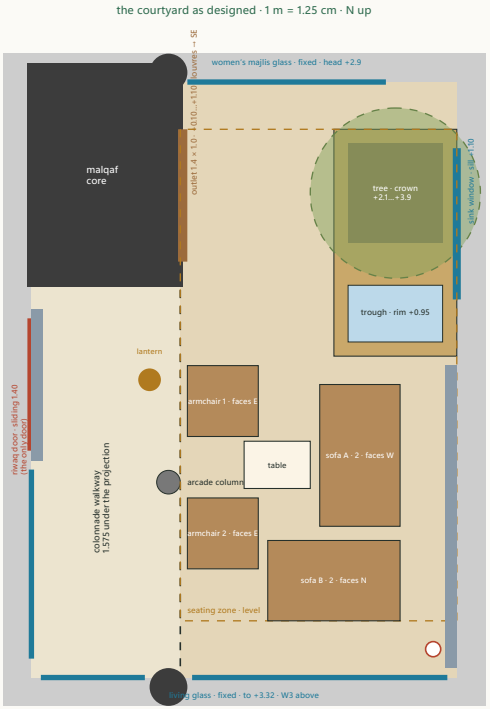
	trough's rim and under the crown	
Malqaf head	1.6 × 2.4 on the roof, x 4.3–5.9 / y 6.4–8.8, openings +11.2…+12.6 on all four faces (3 / 5 slots), the crown +13.2; the W and N faces the catchers; D1 at the shaft's head +7.75, D2 behind the outlet, spring-return, one switch	air_system.typ 27–28, 36, 203
Water + tree enclosure	one box on the kitchen wall x 7.55–8.85 × y 7.1–9.5 (walls 0.15): the planter 1.0 × 1.05 inside, 0.70 deep, rim +0.45 (north, in front of the sink window); the raised trough 1.0 × 0.6 inside, 0.35 deep, rim +0.95, water +0.85, a laminar lip at +1.05 (south); the pump chamber in the divider; both basins tanked, each to the rain system	character.yaml 145–152 · courtyard_layout.typ §3
The tree	**ASSUMED**: **Laurus nobilis** standard or clipped **Ficus nitida**, "a Taif nursery decides"; drawn stem 2.0, crown centre +3.0, radius 0.9 (bottom +2.1, top +3.9); drip irrigation; no root barrier noted	character.yaml 152

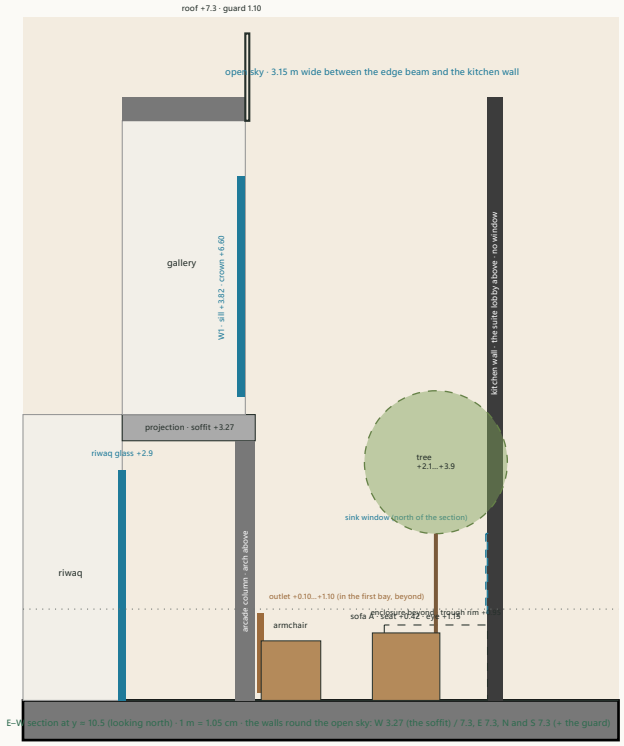
	the thresholds 15 mm at the glass	
Wind	**ASSUMED from the public station**: the warm-season evening wind W ≈ 50 %, NW ≈ 30 %, mean ≈ 21 km/h at 10 m; the house's own wake and the old house west of it estimated, the on-site station still to confirm	malqaf_L5.typ 23–30, 147
Heat and the seating hours	**ASSUMED**: the seating hours 16–23: pleasant and dust-free ≈ 1 800 h / year; > 32 °C in 14–36 % of the hours May–Sep; dust ≈ 60 days / year	malqaf_L5.typ 44–48
Sun	**ASSUMED**: Taif 21.3° N; solar noon ≈ 12:25 local; the sun's altitude at the summer solstice noon 92° − 21.3° ≈ 89° (overhead), at the winter solstice ≈ 45°; Phase 1 computes the shadows	geometry

Every opening onto the courtyard

Opening	What	Where
women's majlis glass (north, y 6.5)	x 6.0–8.1, floor to +2.9, fixed (no door)	presentation.typ 194 · params.yaml 47
riwaq glass door (west, x 4.25)	y 9.1–10.5, sliding 1.40: the only door into the courtyard	presentation.typ 173
riwaq glass (west, x 4.25)	y 10.7–12.7, floor to +2.9, fixed	params.yaml 50
riwaq's first panel (x 4.25, y 6.8–8.9)	the malqaf core's solid rendered face with a built-in bench (no glass since V10.0)	design_gf.typ 100 · PROJECT_V10 §4.63
living glass (south, y 13.0)	x 4.45–5.55 and 6.05–8.75, floor to the slab (+3.32), fixed (no door); W3 above it to the crown +6.60	params.yaml 48–49, 78
kitchen sink window (east, x 8.95)	y 7.3–8.9, sill +1.10, head +2.10; the enclosure in front of it	design_gf.typ 127 · character.yaml 144
malqaf outlet (west, the core's east face x 5.95)	y 7.1–8.5, +0.10…+1.10, timber louvres (free 60 %), the damper D2 behind	air_system.typ 30, 50, 203
malqaf door (the core's south face, y 8.767)	x 4.6–5.6, 1.0 × 2.4, carved timber, onto the covered strip (not the courtyard)	design_gf.typ 100 · air_system.typ 31
gallery windows above (west, x 5.8)	W1-2 y 8.95–10.55 and W1-3 y 11.117–12.717 (1.60 wide, sill +3.82, springing +5.80, crown +6.60); W1-1 (y 6.783–8.383) a blind arch	params.yaml 71–72 · ff_family.typ 124
family living windows above (north, y 6.5)	W2 × 2: x 6.0–7.3 and 7.55–8.85, floor +3.60 to the crown +6.60	params.yaml 76 · ff_family.typ 125
living's upper side (south, y 13.0)	W3: x 6.0–8.8 from +3.32 to the crown +6.60 over the living glass	params.yaml 78
kitchen wall above (east, x 8.95)	the suite lobby's wall, no window to the courtyard (the suite lobby F15 is blind to the court)	rooms_ff.csv, params.yaml windows

Phase 0 · the courtyard as designed: plan (clear faces) and the east–west section through the seating · the summary





Phase 0 · summary

- **The courtyard:** an L of 27.19 m² on axes, ≈ 24.88 m² clear, 4.50 wide between the riwaq glass and the kitchen wall, 6.30 long; the open sky 3.15 × 6.5 m between the projection's edge beam and the kitchen wall, 7.3 m deep (the roof) on three sides and 3.27 m (the soffit) on the colonnade side. The floor level, the porcelain finish, falls of 1 % to the kitchen wall foot.
- **Who opens onto it:** the women's majlis (fixed glass, north), the living (fixed glass to +3.32, south, W3 above), the riwaq (the sliding door 1.40 and fixed glass, west), the kitchen's sink window (east, +1.10…+2.10, behind the enclosure), the gallery's W1-2 and W1-3 (west, above the colonnade), the family living's W2 × 2 (north, above the majlis glass); the suite lobby above the kitchen wall is blind. **One door only** (the riwaq); the malqaf door opens onto the covered strip, not the courtyard.
- **Air:** the malqaf core fills the arcade's first bay; the outlet A low on its east face, 1.4 × 1.0, louvres to the south-east, ≈ 1.79 m³/s at 2.1 m/s in the mean summer-evening wind (W / NW); B would raise and turn it toward the trough. D1 / D2 spring-return on one switch.
- **Water and tree:** one box on the kitchen wall in front of the sink window; the tree north (planter 1.0 × 1.05 × 0.7, rim +0.45), the raised trough south (1.0 × 0.6 × 0.35, rim +0.95, water +0.85, a laminar lip +1.05), a pump in the divider, both basins to the rain system. The species **not decided** (laurel or clipped ficus), the canopy drawn Ø1.8 at +2.1…+3.9.
- **Seating:** six seats with backs (two 2-seat sofas, two armchairs, a table) in the level zone 2.925 × 5.2 (12.09 m² free), facing the outlet arch and the enclosure; no TV.
- **Services:** two slot drains and an overflow to the rain chamber; two lanterns under the soffit; the pump's circuit, sockets and a lighting circuit **not designed**; the chamber's maintenance door on the covered strip.
- **Assumptions to carry:** the wind (station data, the on-site station pending), the tree's size and species, the sun geometry (computed in Phase 1), every air figure (the MEP engineer).

Phase 0 checks passed (the facts read from the files: the courtyard polygon 27.19 m², the zone 12.09 m², the outlet 1.4 × 1, the enclosure 1.3 × 2.4)

Phase 1 · the 22 checks (and the owner's two notes) · pass / fail / unknown, with the numbers

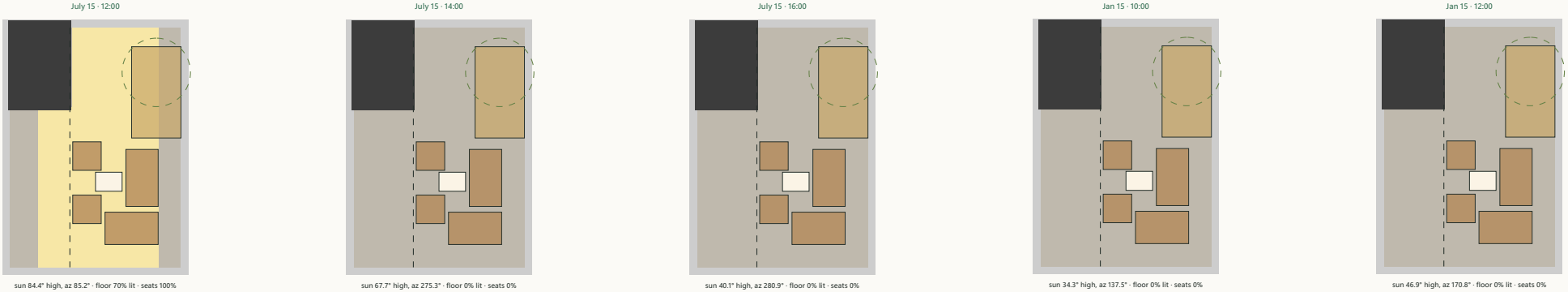
Sun: NOAA solar position for Taif (21.3° N, 40.4° E, UTC+3); a point is in sun when its ray to the sun passes the 3.15 × 6.5 sky opening at +7.3; the seats tested at +0.45 (the cushion), the eye at +1.15. Air: the free-jet estimate of `studies/malqaf_outlet_B.typ` (±30 %). **Amber = assumed.** A = the layout of 10 Nov 2026; B = the outlet study.

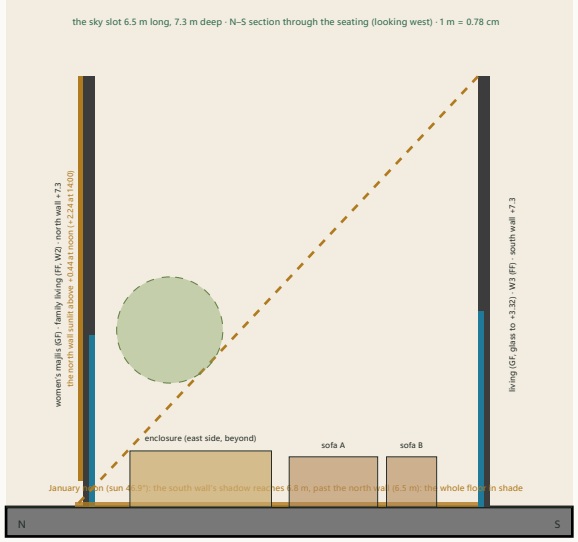
Check	The numbers	Verdict
Air		
1 · the breeze passes over the water and / or under the tree before the seats	A: the outlet +0.10…+1.10, louvres to the south-east: the jet runs over the floor to sofa A and the armchairs; the trough's rim is +0.95 and the crown's bottom +2.1: it crosses neither. B (studied): 59 % of the jet's depth over the water, its top under the crown.	A: FAIL · B: PASS
2 · the seats downstream at seated head height, no jet in anyone's face	A: sofa A faces the outlet 1.95 m away: the jet's top there is +1.44 with ±10° spread, the seated eye +1.15: **the jet is in the face**; armchair 1 at 0.45 m likewise. B: the sofa is 2.0 m off the jet's axis (half-width 1.2): the spill from the kitchen wall only.	A: FAIL · B: PASS
3 · the head faces the summer evening wind	the head's W and N faces catch the W (≈ 50 %) / NW (≈ 30 %) evening wind; the house's own roof wake and the old house to the west assumed until the on-site station.	PASS (wind assumed)
4 · hot air can rise out	the sky opening 3.15 × 6.5 = 20.5 m² over a 24.9 m² floor; the colonnade and the living's double height feed it; the stack ≈ 0.49 m³/s in calm.	PASS
Sun and shade		
5 · shadow on the seating at 16:00 and 18:00 in July, at 10:00 in January	July 16:00 (sun 40° high, WNW): the seats **100% shaded**; 18:00 (13°): 100 % shaded; July 14:00: 100% shaded; July noon (84°): 100% of the seats in sun. January 10:00 (34°, SE): 100 % shaded; January noon (47°): 100 % shaded.	summer PASS · winter FAIL (no sun)
6 · the tree's canopy shades in summer, lets winter sun in	the crown Ø1.8 at +2.1…+3.9 over the planter in the north-east corner: it shades 2.5 m² of the enclosure and the wall, none of the seats (they are 0.3–3 m south-west of it); in winter no sun reaches the floor anyway (5): the canopy is irrelevant to both.	FAIL (no effect)
Water		
7 · moving water, no mosquitoes; heard from the seats; safe for children	a laminar lip +1.05 into the trough (water +0.85): a 0.2 m fall, moving while the pump runs (it must run daily: no standing water). Heard: a laminar sheet is nearly silent (≈ 35–40 dB(A) at 1 m, assumed): not heard from the seats 2.5–4 m away. Safe: a raised trough (rim +0.95) 0.35 deep: a small child cannot fall in; the planter 0.7 deep is soil.	moving PASS · heard FAIL · safe PASS
8 · leaves in the water and their cleaning	the crown's edge (y 8.675) is 0.08 m from the trough (y 8.75): the leaves fall straight into the water; a laurel sheds a little all year, a ficus a lot (and fruit). The trough 1.0 × 0.6 is skimmed by hand; no skimmer or cover.	FAIL
9 · roots vs the water, the drains and the waterproofing	the planter 1.0 × 1.05 × 0.7 (≈ 0.7 m³) sits in a 0.75 recess on the tanked courtyard slab beside the trough and the main drain; a **Ficus nitida** (aggressive, invasive roots) would fill it in 3–5 years, crack it and block its drain: **unsuitable**; a laurel (fibrous roots) stays, but a standard tree in 0.7 m³ stays small (≈ 3–4 m) and needs irrigation every day in summer. No root barrier is drawn.	ficus FAIL · laurel PARTIAL
Use		
10 · every seat sees the tree or the water, not a wall or a service door	sofa A faces west: the core's rendered face and the outlet arch (a wall, 1.5 m away), the enclosure only over its right shoulder; sofa B faces north: the tree and the water ✓; armchair 1 faces east: the enclosure ✓; armchair 2 faces east: the kitchen's plain wall, the enclosure obliquely.	PARTIAL (2 of 4)
11 · conversation: the farthest seat-to-seat < 3.5 m, nobody's back to the group	farthest 2.23 m (sofa B ↔ armchair 1); every piece faces the table; sofa B's back to the living glass, sofa A's to the kitchen wall: the group is a square round the table.	PASS
12 · clearances: the walkway, the doors, the glass cleaning, behind the sofas	the colonnade walkway 1.575 clear (nothing west of the soffit edge); 0.6 at the living glass, 0.5 at the majlis glass, 0.6 behind sofa A at the kitchen wall; the riwaq door's 1.40 opening free; 0.3 only between sofa A and the enclosure (the trough's rim to clean).	PASS (0.3 at the trough: tight)
Comfort and calm		
16 · winter afternoons: sun on the seats, shelter from the cold wind	**no direct sun on any seat on any winter day** (5): the south wall (the living, 7.3 m) shades the whole floor from October to February; at January noon only the north wall's face above +0.55 is sunlit (the women's majlis glass), at 14:00 above +2.32 a seat with its back to the north glass would have its body in the sun for ≈ 2 h round noon. Shelter: the breeze shut by the switch; the well is still air ✓.	sun FAIL · shelter PASS
17 · night use: soft lighting, no glare into the rooms	two hanging lanterns under the soffit, the room lights through the glass; no lighting circuit, no sockets, no fittings in the enclosure or the water designed: unresolved.	UNKNOWN (not designed)
18 · acoustics: the water's sound, the fan's noise, echo	the lip is nearly silent (7); the malqaf's fan behind the core's 0.30 RC wall with its silencer (≈ 38 dB(A) in the chamber): inaudible ✓; the well: 7.3 m of render and glass on four sides over a porcelain floor, ≈ 20 m² of soft surfaces only (the sofas, the tree): a hard, reverberant room (RT ≈ 1.5–2 s): voices ring, the street's sounds are shut out.	fan PASS · water FAIL · echo PARTIAL
19 · scent and planting without mess	one tree only; a laurel gives scent (its leaves) and little mess; a ficus gives none and much mess; no other planting, no climbers on the 7.3 m walls, no pots.	laurel PARTIAL · ficus FAIL
Proportion and beauty		
20 · the proportions (width : height, the sky view), the rhythm, the view from each doorway	the sky slot 3.15 wide between walls 7.3 high: **1 : 2.3** (the courtyard floor 4.5 wide: 1 : 1.6); the traditional comfortable range 1 : 1 to 1 : 1.5. From a seat (eye +1.15) the sky is a patch 29° × 55° (the sky view factor ≈ 0.07 a good courtyard 0.2–0.4): **a deep well**. The rhythm: three arches of 2.17 on the west (the first filled by the core), W1 above them aligned, W2 × 2 north, W3 south: coherent ✓. From the riwaq door you face the kitchen's plain wall 4.5 m away with the enclosure in the corner; from the living's glass you face the majlis glass and the back of sofa B.	FAIL (the well) · rhythm PASS
Build and upkeep		
21 · drainage; power only where needed; the malqaf's maintenance door clear; cleaning the floor, the water and the leaves	the falls, two slot drains, the overflow and the rain chamber ✓; the malqaf door on the covered strip, 1.0 m clear ✓; the level porcelain floor hosed to the drains ✓; the water and the leaves: by hand, daily in autumn (8); power: nothing designed (17).	drains PASS · leaves FAIL · power UNKNOWN
22 · the evaporative cooler option	the space 0.8 × 0.7 × 1.6 in the chamber, its outlet against the louvres, a socket, the floor drain, the water point capped ✓; it would blow through the same louvres (A: into sofa A's face at 1.5–2 m).	PASS (A: the same face problem)
The owner's two notes		
23 · one door only: would a door from the living and / or the women's majlis make the courtyard far better used?	today the living's 3.8 m of glass (x 4.45–5.55, 6.05–8.75) and the majlis's 2.1 m are fixed: from the living's centre the seating is reached in 9.5 m via the riwaq door, 4.9 m through a door in the living's glass. **The living:** the family's main room, double height on this side, the glass to +3.32 under W3: a sliding door 1.40 at x 6.05–7.45 is simple (the frame only; the beam R2 above is untouched), the threshold 15 mm, and the courtyard becomes the living's outdoor room — the strongest single gain found. **The women's majlis:** a door would hand the courtyard to the	living door: FAIL today, a clear gain · majlis door: no

13 · privacy from the guest route and the majlis door	the men's guest route (foyer → majlis) has no view of the courtyard ✓; the **women's majlis** looks straight into the courtyard through 2.1 m of fixed glass (x 6.0–8.1, floor to +2.9): women guests see the family's seating 0.6–6 m away, and the family sees them; no curtain or screen is designed.	guests **PASS** · women's majlis **FAIL**
14 · the guest route never crosses the family seating	the courtyard has one door, from the riwaq (family side); the guests never reach the riwaq.	**PASS**
15 · the views into the courtyard from the living, the women's majlis, the kitchen sink, the gallery	the living (south glass to +3.32): the whole courtyard, sofa B's back in front; the women's majlis (north glass): the sofas' fronts and the core; the sink (window +1.1…+2.1): the trough's rim and the tree, as designed ✓; the gallery (W1-2 / W1-3, sill +3.82): down onto the seating and the enclosure ✓; the family living (W2): down onto the enclosure ✓; the suite lobby: blind.	**PASS** (the living sees a sofa's back)

	women's guests and remove the family's privacy (13): keep the fixed glass with a curtain.	
24 · the "deep well": the sky 3.15 × 6.5 inside 7.3 m walls	proportion 1 : 2.3 and the sky view ≈ 0.07 (20); no winter sun on the floor from October to February (5, 16); summer shade from 13:30 (good); the slot runs north–south, so the only sun that reaches the floor comes near noon. Within the open items the well stays: the walls are rooms (the projection, the family living, the kitchen and the suite lobby above); what can change is how it is lived with: the light colour and the lanterns, a taller tree reaching into the slot, a bright floor, the seating against the north wall for the winter noon, the living's door. Bolder ideas go to Phase 3's "ideas for the owner".	**FAIL** (inherent) · mitigations in Phase 3

Phase 1 · the sun: where the floor and the seats are lit (the sky opening's shadow) · the N–S section at winter noon · the summary





Phase 1 · summary (15 verdicts fail, 16 pass, 2 unknown)

- **The biggest finding is the well.** The sky slot is 3.15 m wide between 7.3 m walls (1 : 2.3; the sky view from a seat ≈ 7 %): **no direct sun reaches the floor or any seat from October to February**; in summer the floor is shaded from about 13:30 (good). The walls are rooms, so the well stays; what the open items can do is live with it: the north wall's face is in the winter noon sun above +0.5, so a seat with its back to the women's majlis glass gets the sun on the body for ≈ 2 h; the light render, the lanterns, a taller tree in the slot, the living's door.
- **The air works but hits the faces (A):** the low outlet aimed at the seats puts the jet's top at +1.44 on sofa A's sitters (eye +1.15) and misses the water and the tree. The raised, turned outlet (B) fixes both: the air over the trough and under the crown, the seats in the spill.
- **The water and the tree are in the wrong relation:** the crown hangs 0.08 m from the trough (leaves in the water), the laminar lip is inaudible, the planter (0.7 m^3) is too small for a ficus (roots, the drains) and small for a laurel; no skimmer, no root barrier.
- **Use:** the conversation square and the clearances are right (11, 12); two of the four seats look at a wall (10); the women's majlis's fixed glass gives no privacy from women guests (13); **the living has no door** onto its own courtyard (23): 9.5 m round by the riwaq against 4.9 m through the glass; a sliding door at x 6.05–7.45 is the clearest single gain; no door from the majlis.
- **Not designed yet:** the lighting and the power (17, 21); the acoustics are hard (18); the planting is one tree (19).
- **Phase 2** ranks these; **Phase 3** draws 2–3 layouts within the open items (the outlet B, the seating turned to the water and the north wall, the enclosure re-made with the tree away from the water, a skimmer, the living's door, the lighting, the floor pattern, the planting) and lists the bolder ideas (the well) for the owner.

Phase 1 checks computed (the sun at 10 moments; the seats' lit shares; the jet geometry; the paths).